

RAIMD: Highly-Reliable Memory Architecture for Heterogeneous Memories

Overview The PIs propose a new memory organization significantly different from the conventional memory organizations designed for DRAMs. Figure ?? contrasts their differences. In the new organization, each memory device (chip) embeds a light-weight memory access scheduling logic to schedule memory operations for itself. The memory controller issues memory access requests, instead of memory operations, to devices. Each device operates independently, yielding high memory concurrency. The interface between the memory controller and memory devices is revised to allow variable access latency. This is a critical difference from conventional memory organizations (such as DDRx) – it allows a memory device to incorporate circuits to improve performance, energy efficiency, reliability, and endurance without violating any timing requirements.

This new organization offers several advantages over the conventional organization. Since a single device serves a memory request, instead of a rank of multiple devices, energy efficiency is significantly improved. Reliability tends to be a challenge with narrow memory ranks – this new organization essentially uses very narrow rank of one device per rank. However, as we will show in our preliminary result, it can actually offer very high reliability with low cost. More importantly, it has excellent extensibility to support new memory technology and device designs, which is critically important to NVMs. The complexity of the internal structures of NVM devices can be hidden behind the new controller/device interface, which allows novel designs for improving NVM performance, energy efficiency, and write endurance. Additionally, it will also help address the major challenges in contemporary DRAM systems, including efficient DRAM refreshing and row-hammer protection, by allowing efficient mechanisms inside DRAM memory devices.

It is not that the new organization lacks its own challenges; however, those challenges can be addressed with technology advances and designs. First of all, the new organization requires high-speed (data rate) memory devices; otherwise, memory latency will increase substantially. At this time, it is no longer a concern. In the near future, the data rate of mainstream DDR5 memory devices will reach 6400 MT/s. Our preliminary result shows that with that data rate, the performance of 8-bit rank size is very comparable to that of 64-bit or 32-bit rank size, and the energy efficiency is much better.

The second challenge is how to support memory error protection with narrow rank. Conventional schemes use (72, 64) ECC words, which does not fit narrow memory ranks. This is a known issue, and one of the PIs has proposed a solution for Mini-Rank [?]. We further propose RAIM-D (Redundant Array of Independent Memory Devices) for SSMD memory system¹. It is similar to the RAID organization for disks, with further optimization to avoid the small write issue of RAID-5. In RAIM-D, each memory device detects error by itself, and the correction is done at system level. This new organization not only support SECDED (single error correcting double error detecting) but also SDDC (single device data correcting). Our preliminary result shows that a SSMD memory system with RAIM-D may incur only xx% of performance overhead but xx% improvement of energy efficiency. The result is consistent with the trend shown by previous studies [?] on Mini-Rank and sub-ranking. With the high rate of today's device, it is now feasible to use a rank as narrow as 8-bit.

The third and the major challenge is to maintain memory channel efficiency while allowing operation timing flexibility. Such a flexibility is the key to utilize diverse memory technologies, or even just to hide the increasing complexities of DRAM devices [?]. In such recent designs (some done by the PIs), a memory device may send a negative feedback upon a memory command if it is not yet ready

¹It is significantly different from previous studies on RAIM [?] (see Section ?? for detail).

for the requested operation, with or without extra feedback about the ready time. Those studies have shown that the approach is effective for DRAM memories, yielding the same level of memory channel efficiency as conventional DRAM designs. However, an advanced approach is needed for NVM and mixed DRAM/NVM memory systems because their latency variant is much higher than DRAM. Such an approach may also support advanced DRAM designs to address DRAM reliability, DRAM refresh bottleneck and row-hammer attack.

Additional information can be embedded within the feedback to help the memory controller retry at a right time. Recent studies [?], including two of the PIs, have shown that high channel efficiency can be achieved in DRAM memory system of this type. However, NVM devices with write leveling circuits and fault tolerance designs may expose much higher timing variations than DRAM. To address this issue, the PIs propose to use perceptron-based neural networks for the memory controller to predict when a memory device will be ready for a given operation. ...

The major challenge of the new organization is that, without rigid timing, the memory controller has to predict the completion time of DRAM operations.

- actually, the latter have a rigid interface such that the memory controller must schedule memory operations with precise timing to ensure correctness.

Equally important, this new organization is much more extensible to NVM (non-volatile memory) devices than the conventional one. NVM devices usually have internal structures, for example write buffers and write-leveling mechanisms, to improve performance and write endurance, and thus are not a good fit for conventional memory organization of a rigid interface.

the conventional, rigid memory organization, and this is very compatible with new types of memory devices, particularly those non-volatile memory devices.

Overview In this project, the PIs will investigate machine learning throttling techniques for heterogeneous memory systems with a memory access protocol of flexible timing. The research extends from preliminary studies of memory access protocols of flexible timing [?], which is the key to utilizing heterogeneous memory technologies including DRAM and NVMs (Non-Volatile Memories).

Unlike a conventional memory system, the memory controller of a VDL memory system does not track the resources and status inside each memory module. It allows interoperability between processors and a wide range of memory modules, including those made by DRAM and NVM technology. This feature is critically important to the adoption of NVM technologies, which have a wide range of timing parameters and may have complex internal structures for performance, write endurance, fault tolerance, and energy efficiency. The preliminary studies of the PIs, as proof of concept, have shown that a VDL protocol performs very comparably to DDRx protocols for DRAM memories. Further research is needed for heterogeneous memories, particularly on those made by NVMs.

In the proposed research, the PIs will study machine learning techniques to improve the efficiency of VDL memory systems for a wide range of memory modules, made by DRAM, NVMs, and with complex internal structures. **In such systems, it is critical to use ML techniques for the memory controller to gauge the internal status of memory modules and then predict the data latencies of pending requests.** A conventional design, which precisely tracks each memory module's internal status, will no longer be feasible because of the mounting complexity. A VDL memory controller, as proposed, uses machine learning to gauge imprecisely the internal status of each memory module and issues memory commands accordingly. It is the imprecise nature of the design that makes it possible for a VDL memory controller to work with a wide range of memory modules, without the need to know precisely their

internal structure and status. Consequently, machine learning plays a critical role in the timeliness of memory commands and thus the efficiency of the VDL protocol.

The PIs further propose a unique design of distributed ML framework for VDL memory systems. In this framework, memory modules provides rich ML feedback per memory command to facilitate machine learning inside the memory controller. Those feedbacks give the memory controller *hints* about the internal status of memory modules that are strongly correlated to memory efficiency. The feedback is memory module specific and thus can be customized by the module designers. Coupled with machine learning techniques, such a memory controller can issue memory commands in an efficient manner without the precise knowledge of memory module internal status. With a high-quality ML model, a VDL memory system may approach or even surpass the efficiency of a conventional memory system of rigid timing, and meanwhile support heterogeneous memory modules.

The feedback includes a field that will be used as a feature for the ML circuit of the memory controller. [Here I must write something about the latency feedback.] With this input, the ML circuit predicts the correct timings of memory command for each memory request. Consequently, the memory controller do not have to issue memory commands with strict timing – it only has to make the best prediction with the help of ML. If a prediction is incorrect and the memory command cannot be served, the memory controller will get a negative feedback and retry the command at a later time. This framework will drastically reduce the complexity of the memory controller: instead of tracking the status of all hardware structures of memory devices, which is infeasible for advanced NVM devices, the memory controller only has to gauge the status using ML modeling.

[In POMI design, the device may send back latency information. This can be added to this protocol: Some binary combinations are reserved for certain feedback, where the others are feature.]

[What feedback? First, tell if an operation can be done or not. Second, Additional information about the feature.] [Scenario 1: Normal memory access, sends precharge, activation, and then data transfer. Device has latches for row address, bank/bank group, and column address. buffers may or may not be ready, because device has some control of the buffers. MC sends row/bank/bg address first, gets a feedback. Then feedback's meaning depends on device. For example, it could be that buffer is ready, not ready; or it can have multiple means, if a NVM device has additional cache/buffer. The feedback may also say the row is not ready and a retry is needed. Then, after some waiting, the MC sends column address to get the data. The feedback first tells if a retry is needed. Additional feedback tells the feature.]

[Draw a diagram here]

[The memory controller still maintains address mapping. That cannot be done using machine learning. ML decides when to send memory command.]

[The feedback should also include information such as if a memory bank is ready for the command and so on]

[Two things to say: Data latency is variable, and the memory controller doesn't have precise knowledge of what's inside a memory module.]

In this research, the PIs will investigate ML techniques to improve the efficiency of variable-latency memory access protocols for heterogeneous memory systems, including DRAMs and various NVMs.

- ML is used to predict when data is ready - Additional information is used to help ML prediction

In the preliminary studies, the PIs have proposed new memory access protocols with carefully relaxed timing requirements, which is necessary for heterogeneous memory systems. [Define heterogeneous memories?] In the proposed research, the

A key challenge in this approach is to address a potential loss of efficiency from the flexibility of the

memory access protocol. In this research, the PI proposes new ML throttling techniques to improve the efficiency. [Define the goal of the research: Very high efficiency, complexity]

The complexity of memory subsystem has been increasing substantially to meet the ever-increasing demands for higher capacity, bandwidth, throughput, reliability, and energy efficiency. Furthermore, non-volatile memories are poised to challenge DRAM, the dominant technology for the past decades. This transition will have far-reaching impacts on computer systems design [expand to OS, security, and so on]. However, it also substantially increases the complexity of memory subsystem design, which is under high pressure to meet the increasing demands from data-intensive workloads and must balance between performance, energy efficiency, and reliability.

[Need to revise this part ...]

A core issue to this problem is the rigid design of existing memory access protocols, with DDRx protocols being the dominant design. A memory access protocol articulates how a memory controller communicates to memory devices, e.g., DDR4/DDR5 DRAM devices (chips). The current DDRx protocols evolve from the SDRAM (synchronous DRAM) from 1990s [double check the date], with drastic improvements of data rate. However, like in SDRAM, they are still rigid protocols, in which memory devices are slave devices that respond to memory commands, e.g., precharge, row activation, and data read/write, in a very strict timing requirements. [Give an example of timing.] The strict timing requirement was necessary, in the era of DRAM memory, to minimize the cost of DRAM devices and meanwhile maintain the correctness and efficiency of memory bus transactions.

However, in the new era of heterogeneous memories, the strict timing requirement has become a severe constraint. It leaves no room for a memory device to adopt any design improvements that affect the latency of memory operations. For example, [PCM write leveling, rowhammer attack, and so on.] Thus, many creative design ideas proposed so far may not be adopted because they will violate the strict timing requirements.

A [add cap] flexible access protocol for main memory, however, must make very careful trade-offs between flexibility and efficiency. Otherwise, it may lead to significant loss of performance or energy efficiency [I want to talk about latency and bandwidth]. For example, ...

The PIs will investigate ML-throttled memory access scheduling for memory access protocols with relaxed timing requirements. The PIs have studied such protocols in previous studies [?], for DRAM and NVM [double check on NVM], and found that they can perform comparably or even better than conventional

The PIs have found in a preliminary study [?] that ML is very promising in reducing the inherent overhead of a flexible memory access protocol... [Write about preliminary results]

The preliminary study was based on DRAM devices. Research on a flexible memory access protocol for heterogeneous NVM systems is much more challenging, and much more important, because of diversity of NVM devices. ... [Discuss timing variants, write leveling, error tolerance, ECC, but avoid redundancy.]

References